FORMATIVE ASSESSMENT – 1 (FA-1)

MATHEMATICS • CLASS 9 • SLIP TEST

Question 1

Find three rational numbers between **3 / 5** and **4 / 5** using equivalent fractions.

Question 2

Rationalize the denominator and simplify the expression: **(√5 + √3) / (√5 − √3)**.

Question 3

Evaluate the value of the polynomial $\mathbf{p(x) = 5x - 4x^2 + 3}$ at **x = 0**, **x = −1**, and **x = 2**.

Question 4

Factorise the quadratic polynomial $\mathbf{12x^2 - 7x + 1}$ by splitting the middle term.

Question 5

Evaluate the product **103 × 107** directly without multiplying, using a suitable algebraic identity.

SOLUTIONS & SCHEME OF WORK

CLASS 9 MATHEMATICS • FA-1 SLIP TEST

Solution 1

To find 3 rational numbers, multiply numerator and denominator by (3 + 1) = 4:

3 / 5 = (3 × 4) / (5 × 4) = **12 / 20**

4 / 5 = (4 × 4) / (5 × 4) = **16 / 20**

Three rational numbers lying between 12/20 and 16/20 are:

Ans: 13 / 20, 14 / 20 (7 / 10), 15 / 20 (3 / 4)

Solution 2

Multiply numerator & denominator by the conjugate of (√5 − √3), which is (√5 + √3):

= [(√5 + √3)(√5 + √3)] / [(√5 − √3)(√5 + √3)]

= $[(\sqrt{5})^2 + 2(\sqrt{5})(\sqrt{3}) + (\sqrt{3})^2]$ / $[(\sqrt{5})^2 - (\sqrt{3})^2]$ = [5 + 2√15 + 3] / [5 − 3]

= [8 + 2√15] / 2 = 2(4 + √15) / 2

Ans: 4 + √15

Solution 3

Given polynomial: **$p(x) = 5x - 4x^2 + 3$**

• At **x = 0**: $p(0) = 5(0) - 4(0)^2 + 3 = 0 - 0 + 3 = \mathbf{3}$

• At **x = −1**: $p(-1) = 5(-1) - 4(-1)^2 + 3 = -5 - 4(1) + 3 = -9 + 3 = \mathbf{-6}$

• At **x = 2**: $p(2) = 5(2) - 4(2)^2 + 3 = 10 - 4(4) + 3 = 10 - 16 + 3 = \mathbf{-3}$

Ans: p(0) = 3, p(−1) = −6, p(2) = −3

Solution 4

Given: **$12x^2 - 7x + 1$**

Find two numbers whose product is 12 × 1 = 12 and sum is −7. The numbers are **−4** and **−3**:

= $12x^2 - 4x - 3x + 1$

= 4x(3x − 1) − 1(3x − 1)

Ans: (3x − 1)(4x − 1)

Solution 5

Write as (100 + 3)(100 + 7) and use the identity **$(x + a)(x + b) = x^2 + (a + b)x + ab$**:

Here, x = 100, a = 3, and b = 7:

= $(100)^2$ + (3 + 7)(100) + (3 × 7)

= 10,000 + 10(100) + 21 = 10,000 + 1,000 + 21

Ans: 11,021